

Chapter 10: Data Handling and True/False (Statistics and Probability)

Worksheet 10: Statistics and Introduction to Probability

Learning Objective: Students will organise data in frequency tables and histograms; calculate mean, median, and mode for grouped and ungrouped data; understand basic probability (experimental and theoretical); and compute probability of simple events.

Worked Example

A coin is tossed 50 times: Heads = 28, Tails = 22.

Experimental $P(\text{Heads}) = \frac{28}{50} = 0.56$

Theoretical $P(\text{Heads}) = \frac{1}{2} = 0.5$

Mean of grouped data: Class 0–10 (freq 5), 10–20 (freq 8), 20–30 (freq 12), 30–40 (freq 5): Midpoints 5, 15, 25, 35.

Mean = $\frac{5 \times 5 + 8 \times 15 + 12 \times 25 + 5 \times 35}{30} = \frac{25 + 120 + 300 + 175}{30} = \frac{620}{30} \approx 20.7$

Questions 1–5: Easy / Recall

Q1. A die is rolled. What is the probability of getting: (a) a 4, (b) an even number, (c) a number less than 3?

Q2. The probability of an event is 0.35. What is the probability of it NOT occurring?

Q3. Find the mean of: 12, 18, 25, 30, 15.

Q4. Find the median of: 7, 3, 10, 5, 8, 2, 9.

Q5. Find the mode of: 4, 6, 4, 8, 6, 4, 7, 6, 4.

Questions 6–10: Basic Application

Q6. A bag has 5 red, 3 blue, and 2 green balls. One is drawn at random. Find $P(\text{red})$, $P(\text{blue})$, $P(\text{not green})$.

Q7. Two coins are tossed. List all outcomes. Find $P(\text{both heads})$, $P(\text{at least one tail})$.

Q8. The heights (in cm) of 10 students: 150, 155, 148, 162, 155, 170, 148, 155, 160, 152. Find mean, mode, and median.

Q9. A card is drawn from a pack of 52. Find: $P(\text{an ace})$, $P(\text{a red card})$, $P(\text{a king of spades})$.

Q10. 30 students took a test. Mean score = 65. Later, it was found one score was misread as 45 instead of 54. Find the correct mean.

Questions 11–15: Practice and Skill Building

Q11. Frequency table:

Marks	Frequency
0–10	4
10–20	7
20–30	12
30–40	10
40–50	7

Find the estimated mean. Which is the modal class?

- Q12.** A die is thrown twice. Find the probability that: (a) both show 6, (b) sum is 8, (c) product is 12.
- Q13.** The mean of 20 numbers is 45. If two numbers 60 and 80 are removed, find the new mean.
- Q14.** In an experiment, a drawing pin was tossed 200 times: point up = 120, point down = 80. Estimate the probability of “point up.” If tossed 500 times, how many times would you expect “point up”?
- Q15.** Describe the difference between experimental probability and theoretical probability. Give an example of each.

Questions 16–18: Higher Order Thinking

- Q16.** Letters of the word MATHEMATICS are written on cards and placed in a box. A card is drawn at random. Find $P(\text{a vowel})$, $P(\text{the letter M})$, $P(\text{a letter that appears more than once})$.
- Q17.** The mean of 50 observations is 36. The mean of first 25 is 32 and the mean of last 25 is x . Find x .

- Q18.** Class scores (out of 100) for 40 students: below 40 = 5, 40–60 = 12, 60–80 = 18, above 80 = 5. A student is chosen at random. Find $P(\text{below } 40)$, $P(60\text{--}80)$, $P(\text{passed if pass mark is } 40)$.

Questions 19–20: Real-Life Applications

- Q19.** Record your class's marks in the last Maths test. Organise into a frequency table with class intervals of width 10. Draw a histogram. Calculate the mean, median class, and modal class. (Teacher Verification Required)
- Q20.** A weather report says the probability of rain tomorrow is 0.7. Raju says "It will definitely rain." Priya says "It will probably rain but not certainly." Who is correct? If it rains 7 months out of 12 in your city, what is the experimental probability of rain in a randomly chosen month?

Reflection Question: Can the mean of a data set be a value that doesn't appear in the data set? Give an example. Can the mode appear more than once?